typically are designed for data that fits in physical memory. Consider that you are given two matrices A and B of dimensions MXN and NXR. The matrix dimensions are such that neither will fit in the main memory. You are asked to write an algorithm for multiplying the two matrices and storing the result into a matrix C of dimensions MXR on disk. Can you outline the approach you will take for minimising the number of disk accesses that your *matrix_multiply* function will perform? Now you are told that matrices A and B are such that most of their elements are zero. You are again asked to compute the matrix multiplication of these two matrices. Can you come up with an algorithm that minimises the number of disk accesses?

5. You have been given the problem of sentiment analysis, where given a sentence, you need to classify the sentence as positive, negative or neutral sentiment bearing. You have been given large quantities of labelled data of example sentences with their target sentiment polarity annotated. Given that there can be words in the test input sentences that are not present in the training set examples, how will your approach handle the out-of-vocabulary words in these sentences? If you decide to ignore the out-of-vocabulary words, what will be the impact on the test set classification accuracy? Can you devise a better approach to handling this issue?
6. You have been given 5-year data of the daily stock price of a company, and asked to predict its stock price at an interval of 30 days for the next one year. How will you go about modelling this problem with a traditional machine learning based solution? What is the type of classifier you will use? What will be the input feature vector? Instead of using a traditional machine learning solution, if you are asked to use a neural network based model, what type of neural network will you use?
7. You are working on developing a part of a large image classification system of indoor household images taken from different cameras placed inside the house. Such a system can be part of a smart home project or for home security considerations. Since many of your customers have pet cats, your system should be able to recognise them. You have designed a machine learning system that can recognise cat images from non-cat images. Note that the household cameras being used in the set up are quite cheap, and can end up producing poor quality images due to variations in lighting, camera motion, etc.

You decide to train your system using the zillions of pet images from the Internet, which are typically high quality. You download 100,000 labelled pet images for training your cat recognition classifier. You also obtain 5000 home-camera generated images, which can include cat images as well. You plan to use 2500 of these 5000 images as a validation set for tuning the hyper-parameters. The remaining 2500 images will be used as a test set for final evaluation.

After you have finished developing the initial model and are about to start the hyper-parameter tuning exercise as the next step, your friend says that your methodology is incorrect. This is because your training set and validation set come from different data distributions and hence it is likely that your chosen model will perform badly on the test set. Do you agree with him? If no, explain why. If yes, explain how will you change your methodology? (Note that you cannot increase the indoor household camera image data, which is only 5000 images, of which a few hundred are cat images.)

8. You have been asked to set up a deep learning system that can recognise faces of employees and send signals to open the entrance gate to the laboratory, as the employee approaches the gate. You have heard a lot about the advantages of end-to-end deep learning.

In this case, if you decide to go with the end-to-end deep learning approach, you only provide the image of the employee approaching the door and the neural network provides an open/close signal output which can be used to drive the entrance gate. The alternative is to employ a pipeline approach, wherein the first phase identifies the 'face part' from the image and puts a bounding box around it. The next phase does the face match recognition to check if he/she is a valid employee of the company. Which approach will you choose? Explain the rationale behind your choice.

9. Given that deep neural networks need large amounts of training data to generalise well, can you explain methods by which you can artificially increase the size of your training data? (*Clue:* Think of transformations such as scaling, rotation and translation. While it may be obvious to employ these transformations in the case of image data sets to increase the size of training data, what kind of transformations can you employ in case of text data?)
10. How can you compare the performance of two different machine learning algorithms? For example, consider that you have been given two learners – one based on support vector machines (SVMs) and another using neural networks for classifying documents. Let us also assume that SVM based learner performs better than the neural network based learner when the training data size is M, whereas neural network performs better than SVM when the training data size is much less than M, but is poorer in performance by a few points when comparing training data sizes of M and above. Which training algorithm do you prefer and can you explain the rationale behind your choice?

Feel free to reach out to me over LinkedIn/email if you need any help in your coding interview preparation. If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Wishing all our readers happy coding until next month! Stay healthy and stay safe. END

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